Geodesy and Geodynamics 14 (2023) 551–558

Contents lists available at ScienceDirect

Geodesy and Geodynamics

journal homepage: http://www.keaipublishing.com/geog

A new method for coseismic offset detection from GPS coordinate time series

Zhiwei Yang [a], Guangyu Xu [a, b, c, *], Tengxu Zhang [d, e], Mingkai Chen [a], Fei Wu [a], Zhiping Chen [a]

[a] School of Surveying and Geoinformation Engineering, East China University of Technology, Nanchang 330013, China
[b] Key Laboratory of Mine Environmental Monitoring and Improving around Poyang Lake of Ministry of Natural Resources, East China University of Technology, Nanchang 330013, China
[c] Key Laboratory for Digital Land and Resources of Jiangxi Province, East China University of Technology, Nanchang 330013, China
[d] Hubei Luojia Laboratory, Wuhan 430079, China
[e] College of Resources and Environmental Science and Engineering, Hubei University of Science and Technology, Xianning 437100, China

ARTICLE INFO

Article history:
Received 4 July 2023
Received in revised form
10 September 2023
Accepted 11 September 2023
Available online 5 October 2023

Keywords:
GPS
Coordinate time series
Coseismic offset
Signal detection

ABSTRACT

Currently, the extraction of coseismic offset signals primarily relies on earthquake catalog data to determine the occurrence time of earthquakes. This is followed by the process of differencing the average GPS coordinate time series data, with a time interval of 3 to 5 days before and after the earthquake. In the face of the huge amount of GPS coordinate time series data today, the conventional approach of relying on earthquake catalog data to assist in obtaining coseismic offset signals has become increasingly burdensome. To address this problem, we propose a new method for automatically detecting coseismic offset signals in GPS coordinate time series without an extra earthquake catalog for reference. Firstly, we pre-process the GPS coordinate time series data for filtering out stations with significant observations missing and detecting and removing outliers. Secondly, we eliminate other signals and errors in the GPS coordinate time series, such as trend and seasonal signals, leaving the coseismic offset signals as the primary signal. The resulting coordinate time series is then modeled using the first-order difference and data stacking method. The modeling method enables automatic detection of the coseismic offset signals in the GPS coordinate time series. The aforementioned method is applied to automatically detect coseismic offset signals using simulated data and the Searles Valley GPS data in California, USA. The results demonstrate the efficacy of our proposed method, successfully detecting coseismic offsets from vast amounts of GPS coordinate time series data.

© 2023 Editorial office of Geodesy and Geodynamics. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

1. Introduction

GPS techniques have been extensively employed in geophysical research, encompassing areas such as plate motion, earthquake observation, and early warning systems. This is primarily due to the capability of GPS to provide complete and continuous time series data for obtaining accurate surface deformation measurements [1,2]. The GPS coordinate time series obtained through continuous observations effectively capture various stages of the seismic cycle, including interseismic deformation, coseismic deformation, and postseismic deformation processes [3]. The coseismic offset can be regarded as a quasi-static adjustment of the crustal strain during an earthquake [4]. When an earthquake occurs, the energy released from the rupture propagation and the receivers located within the epicenter area can detect this energy propagation [5]. The analysis of coseismic offsets recorded by GPS receivers facilitates the investigation of earthquake source mechanisms during seismic events. This exploration holds significant value for regional tectonic studies. The detection of coseismic offset signals in GPS coordinate time series plays a pivotal role in studying surface deformation and understanding the physical mechanisms underlying earthquakes. Furthermore, accurate extraction of coseismic offsets enables the

* Corresponding author. School of Surveying and Geoinformation Engineering, East China University of Technology, Nanchang 330013, China.
E-mail address: gyxu@ecut.edu.cn (G. Xu).

Peer review under responsibility of Institute of Seismology, China Earthquake Administration.

Production and Hosting by Elsevier on behalf of KeAi

https://doi.org/10.1016/j.geog.2023.09.001

1674-9847/© 2023 Editorial office of Geodesy and Geodynamics. Publishing services by Elsevier B.V. on behalf of KeAi Communications Co. Ltd. This is an open access article under the CC BY-NC-ND license (http://creativecommons.org/licenses/by-nc-nd/4.0/).

construction of more reliable models for GPS coordinate time series, leading to the generation of more precise GPS velocity fields. This approach establishes a more credible foundation for studying crustal deformation, stress, and strain conditions, as well as earthquake occurrences [6]. Currently, limited research is available on the automatic extraction of coseismic offsets from a considerable volume of GPS coordinate time series data without relying on external earthquake catalog data.

Global position system (GPS) coordinate time series are an important source of geophysical data, typically including tectonic and non-tectonic deformation, residual atmospheric delay signals, observation errors. As a significant tectonic deformation signal recorded in GPS coordinate time series, the coseismic offset can be used to study surface deformation, constrain coseismic fault parameters, and invert coseismic slip distribution. And the coseismic offset provides insights into strain energy release and changes in the crustal strain field during an earthquake. The coseismic offset should not be interpreted as reaching a certain compression strain limit or plastic deformation stage but rather as a quasi-static adjustment during an earthquake. Currently, there is limited research on detecting coseismic offset signals in GPS coordinate time series due to the difficulty in identifying offset signals with unknown occurrence times. This difficulty often results in false positive or missed detections [7]. Therefore, manual identification remains the primary method for detecting these coseismic offset signals. In order to detect coseismic offset signals, it is commonly practical to examine the coordinate time series of all GPS sites within the assumed source range area, which helps determine whether there is a true coseismic offset signal in the GPS coordinate time series. The current methods for detecting coseismic offset signals heavily rely on manual identification of the original GPS coordinate time series observations [8]. However, the manual detection method is time-consuming and laborious, and even less able to face massive data processing. To address this issue, we propose a new comprehensive detection and identification method, which has been experimentally validated in terms of accuracy and detection range. The GPS coordinate time series contains numerous signals and errors, so it is necessary to remove other large signals and establish GPS coordinate time series model in order to extract the desired coseismic offset signals. Given that the disparity between the front and back of the GPS coordinate time series data is not large, first-order difference and data stacking are applied to highlight the offset signals to achieve automatic detection of the coseismic offset signals in the epochs recorded at GPS sites. Firstly, the effectiveness of the proposed method is validated through a set of simulated experiments involving presupposed earthquake occurrence times. Then, the correctness of the method is tested using earthquake data from the Searles Valley earthquake in California, USA, provided by the Nevada Geodetic Laboratory.

2. Methodology

2.1. Removing long-term trend and seasonal components from GPS coordinate time series

The initial GPS coordinate time series consists of various components, including site velocities, offsets due to antenna movements or equipment replacements, seismic deformations, seasonal variations for the annual and semiannual cycles, as well as transient deformations and observation noise that are not modeled [9–11]. To accurately detect coseismic offset signals in subsequent analysis, it is necessary to model the GPS coordinate time series. Due to the low correlation between components in different epochs [12], the GPS coordinate components (ΔE, ΔN, ΔU) at epoch t_i can be modeled as follows [13]:

$$y(t_i)=a+bt_i+c\sin(2\pi t_i)+d\cos(2\pi t_i)+e\sin(4\pi t_i)+f\cos(4\pi t_i)+\sum_{j=1}^{n_g}g_jH\left(t_i-T_{g_j}\right)+\sum_{j=1}^{n_k}k_j\left(1-e^{\left[-\left(\frac{t_i-T_{kj}}{\tau_j}\right)\right]}\right)H\left(t_i-T_{k_j}\right)+\varepsilon \tag{1}$$

where t_i represents the epoch starting at t_0 in years; a represents the initial displacement value; b represents the interseismic velocity; c, d, e, and f represent the coefficients for the annual and semiannual cycle terms; and g represents the coseismic offsets in the T_g epoch. The exponential model based on the linear viscoelastic rheological mechanism of the crust $k\left(1-e^{\left[-\left(\frac{t_i-T_k}{\tau}\right)\right]}\right)$ represents the postseismic deformation with an amplitude of k and an attenuation constant of τ at the T_k epoch [14]; ε is the observation error; and H represents the Heaviside step function [13]:

$$H=\begin{cases}0, t_i-T<0\\ 1, t_i-T\geq 0\end{cases} \tag{2}$$

The parameters a and b in the above model represent the site position and the linear velocity of the site, namely the mean and trend term of the GPS coordinate time series, which primarily result from the relative motion between tectonic plates [15]; (semi) annual cycle terms, also known as the seasonal terms, are mainly caused by the periodic variation of groundwater levels [16]; $\sum_{j=1}^{n_g}g_jH(t_i-T_{g_j})$ is the offsets correction term, T represents the occurrence time of the offset signals, the step function H is 0 before the offset and 1 after the offset; the exponential model represents the post-earthquake rate decay after the occurrence of the earthquake. This model is extensively utilized in GPS coordinate time series analysis, and it can remove the influential signals and errors in the original coordinate time series, which plays an important role in accurately identifying and detecting coseismic offset signals.

2.2. Automatic detection and extraction method of coseismic offset signals from GPS coordinate time series

The detection of coseismic offset signals in GPS coordinate time series commences with the processing and analysis of time series data from a large number of sites. To ensure the reliability of the analysis, the data is filtered based on the amount of available epoch data to select sites with relatively complete coordinate time series. Then, an automatic threshold is set based on the observed data, and if the difference of adjacent data exceeds the threshold, it is considered an outlier. In such instances, it becomes essential to remove the outlier data and employ the moving average method for interpolation to fill the gaps and restore the smoothness of the time series. After the completion of the data pre-processing stage, the detection procedure of coseismic offset signals is carried out. The initial step is to determine the optimal attenuation constant τ using the trial-and-error method [17,18], and then use τ as a known value to determine the other parameters using the least squares method and GPS coordinate time series model. Subsequently, the trend and seasonal signals presented in the GPS site displacement time series are removed. And the first-order difference and data stacking processing are performed for all time series, to achieve automatically detecting the coseismic offset signal in GPS coordinate time series.

During an earthquake, GPS sites are affected by the earthquake rupture, leading to noticeable location changes that can be recorded in the GPS coordinate time series. In this study, several steps are taken to automatically detect coseismic offset signals. First, outliers in the time series data are removed, and then select the

weighted average of the first-order difference values as its threshold. By comparing the values of the first-order differences of the GPS coordinate time series with this threshold, larger differences indicate larger anomalies. To enhance the visibility of the coseismic offset signals, signal amplification through data stacking is employed. Therefore, we primarily employ the first-order difference and data stacking to automatically detect the coseismic offset signals in the GPS coordinate time series. Some GPS sites may display non-linear factors that are not accounted for in the models and can be ascribed to external forces or non-seismic influences. To ensure that these non-linear factors do not affect the detection of seismic coseismic offset signals, it becomes crucial to validate the consistency between coseismic offset and offsets obtained from various sources. This validation process significantly enhances the reliability of the automated detection. Consequently, this automated detection method compares coseismic offset signals and the values of first-order differences against the established threshold.

3. Synthetic tests

We used synthetic GPS coordinate time series to assess the feasibility of the proposed method. Taking 63 GPS sites in a certain area of the Nevada Geodetic Laboratory GPS Networks (Fig. 1), we extracted long-term linear trend and seasonal terms from the original GPS coordinate time series of these sites. Subsequently, we simulated coseismic offsets caused by two earthquakes using the elastic dislocation model [19] for these sites. These coseismic offset signals were then combined with the epochs of the original GPS coordinate time series and the coefficient of linear trend and seasonal terms to generate simulated GPS coordinate time series based on Eq. (1). Then we introduced Gaussian noise into the simulated data as the observation errors. Finally, our proposed method detects the coseismic offsets in the simulated GPS coordinate time series. An important aspect of the coseismic offset signal detection method in this study is automatic to select different threshold values. Since each site is affected differently by the earthquake, the choice of threshold values varies accordingly. Considering the long observation period, using a simple arithmetic mean as the threshold value for a given data set is inadequate. Therefore, a weighted average is employed as the threshold value, with the weights determined by the magnitudes of the observations, which ensures that each GPS site has its independent threshold value. Moreover, the threshold is not determined solely by data in one direction of the GPS coordinate time series but by the weighted average of the GPS coordinate time series data in the East and North directions.

Taking the PILO site as an example, Fig. 1 shows the station location and simulated time series of the PILO site. Using the proposed method, the time series of 63 GPS sites were modeled. Fig. 2 shows the original GPS displacement time series of partial sites. Fig. 3 presents the detrended GPS displacement time series obtained by removing the trend and seasonal components. As shown in Fig. 4, all GPS displacement time series, after removing the trend and seasonal components, were stacked together to make the coseismic offset signals obvious. As seen in Fig. 2, it can be observed that the complete GPS coordinate time series contains obvious trend and seasonal terms, while the detrended data shows a stationary time series, reducing the influence from these two components. This removal of the two main components (trend and seasonal terms) in the GPS displacement time series plays an important role in the accurate detection of coseismic offset signals. The results show that the coseismic offset signals of the simulated GPS coordinate time series appear in the epochs of 2019.71 and 2021.12, respectively, consistent with our presupposed time in simulation. Table S1 presents the coseismic offsets of 6 selected sites exhibiting large coseismic offsets. The table includes the coseismic offsets simulated in the original data and the coseismic offsets determined using our proposed method, corresponding to the two occurrence times of the coseismic offset signals.

4. Application to GPS coordinate time series of Nevada Geodetic Laboratory

For this study, we utilize coordinate time series of 200 continuous GPS stations in Searles Valley area processed by the Nevada Geodetic Laboratory (NGL). The objective is to detect coseismic

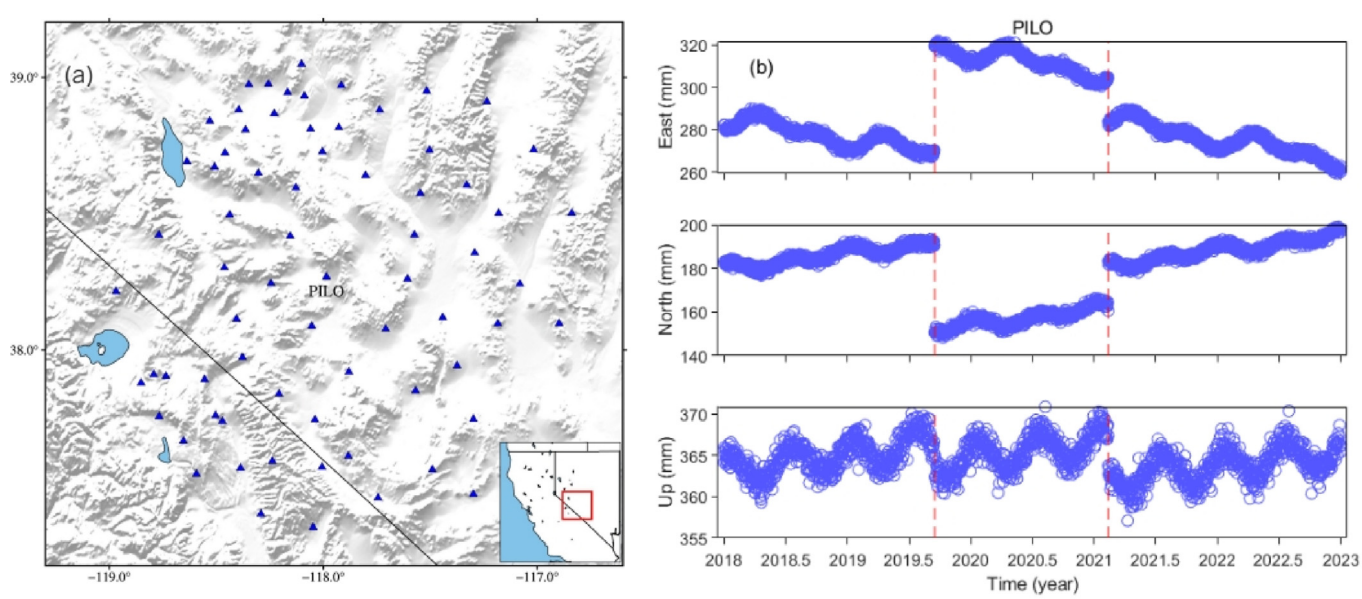

Fig. 1. (a) Locations of GPS stations (blue triangle) used in synthetic tests. (b) Simulated GPS coordinate time series of PILO site. The purple dots represent the observed values, and the red dashed line represents the presupposed occurrence time of the offset signal.

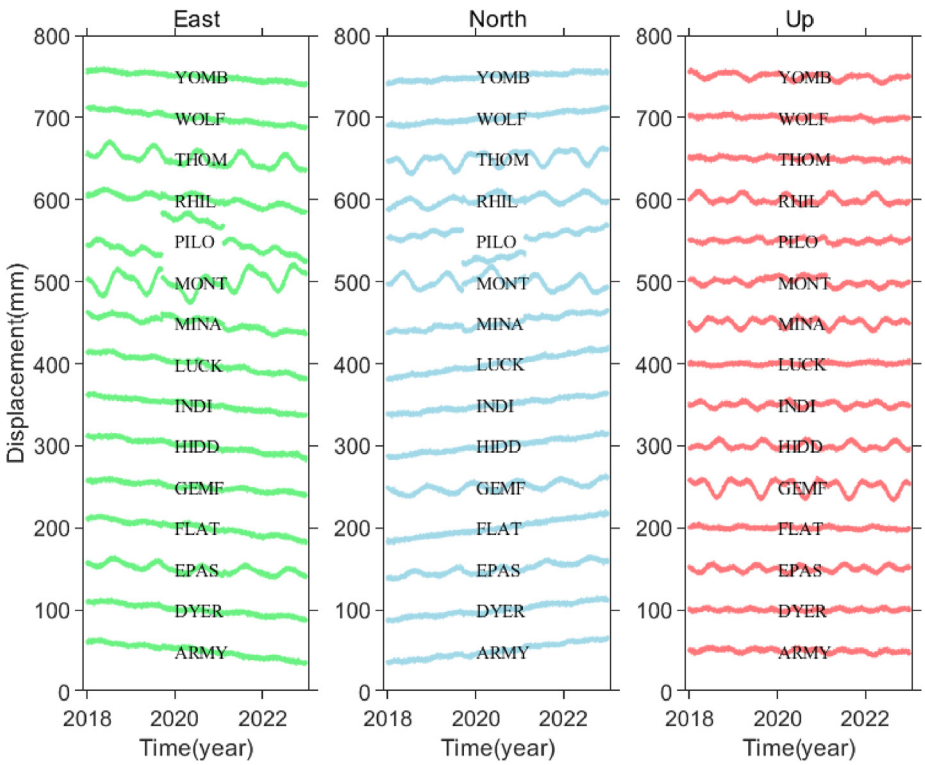

Fig. 2. Complete GPS coordinate time series of 15 sites out of all 63 sites. The green dots represent the simulated time series in the East direction, the light blue dots represent the simulated time series in the North direction, and the red dots represent the simulated time series in the Up direction.

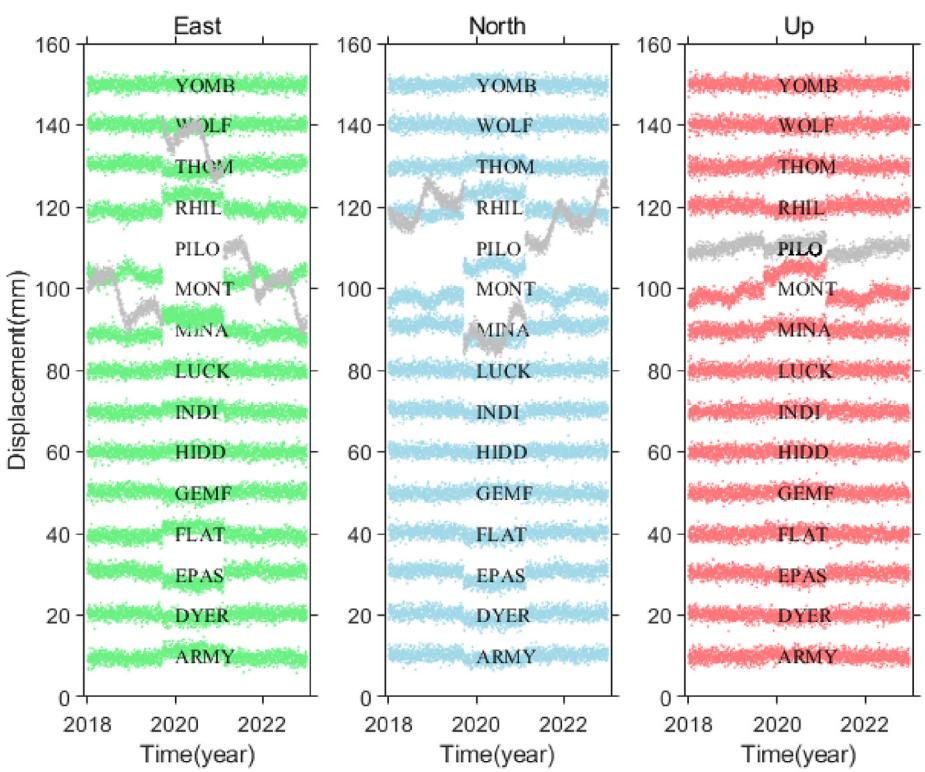

Fig. 3. The detrended GPS coordinate time series of 15 sites out of all 63 sites. The green dots represent the detrended time series in the East direction, the light blue dots represent the detrended time series in the North direction, and the red dots represent the detrended time series in the Up direction. And the gray dots represent the time series of the PILO site.

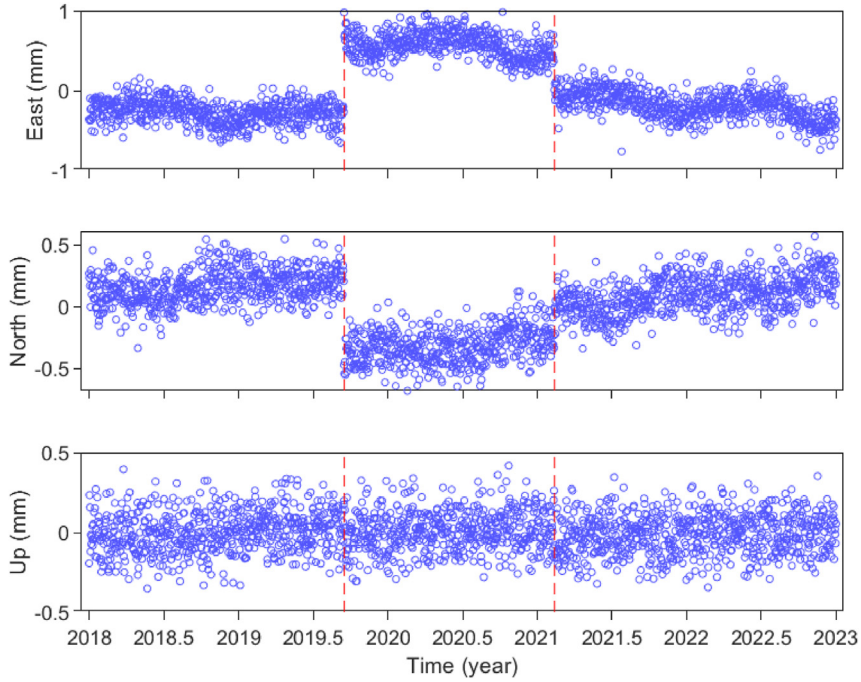

Fig. 4. Stacking of all the detrended GPS coordinate time series of 63 sites. The purple dots represent the stacking values, and the red dashed line represents the detected occurrence time of the coseismic offset signal.

offset signals in the GPS displacement time series data. To ensure the accuracy of signal detection, sites with data coverage less than 80% were eliminated from the analysis. As a result, 164 sites with sufficient data coverage were considered for further analysis. Due to the unknown earthquake event, it is possible to observe different coseismic offset signals at various sites. Moreover, the distance between the sites and the epicenter of the earthquake is also unknown, implying that some sites may not exhibit any coseismic offset signals.

After excluding sites with poor data quality, the GPS displacement time series are modeled with Eq. (1). Firstly, we used the trial-and-error method to determine the parameter τ. Fig. S1 shows that parameter $\tau = 0.8$ provides the minimum Root Mean Squared Error (RMSE) value, which suggests that the optimal attenuation constant is 0.8. Subsequently, the other parameters are determined by the least squares method. Fig. 5 represents the original GPS displacement time series of 16 sites out of all 164. Fig. 6 shows the detrended GPS displacement time series after removing the trend and seasonal components. In Fig. 7, all detrended GPS displacement time series are stacked to make the coseismic offset signals clear. When selecting the attenuation constant, various numerical values are used to simulate the model, and the suitable decay constant is chosen based on RMSE values. Through the aforementioned detection method, the occurrence of a coseismic offset signal in the GPS coordinate time series is detected to be 2019.51. As shown in Fig. 6 and Fig. S2, the detected occurrence time is consistent with the coseismic offset seen in some individual GPS displacement time series, which suggests that the automatic detection method is feasible and the detection result is accurate.

5. Results

The proposed method effectively detected the coseismic offset signals of the simulated GPS coordinate time series occurring in the epochs of 2019.71 and 2021.12. In the actual observation, the coseismic offset signal was observed to occur in 2019.51. By referring to available information, it is known that a magnitude 7.1 earthquake took place in Southern California, USA, on July 6, 2019, at 3:19:53 a.m. UTC time. The epicenter of the earthquake was situated approximately 18 km northeast of the town of Ridgecrest in Southern California (35.770°N, 117.599°W), with a source depth of about 8 km. The results indicate that the BEPK site, located 44.5 km from the epicenter, was significantly affected by this earthquake, consistent with the information obtained from the references. Overall, the method proposed in this paper is accurate and efficient, which can avoid relying on earthquake catalog information and the tedious and complex manual detection process of the coseismic offset signals. Finally, the automatic detection and extraction of the coseismic offset signal from the GPS coordinate time series is realized. The determination of the attenuation constant is a key aspect of the method described in this paper. It entails the search for an optimal value that minimizes the RMSE value from a vast set of candidates.

6. Discussion

The automatic detection method of coseismic offset signals presented in this study relies on established mathematical models and the idea of observation stacking for improving signal-to-noise ratio. It can accurately detect the occurrence of coseismic offset signals in the recorded time series data of GPS sites. However, further investigation is required to determine the accuracy of detecting deformation and coseismic offset signals that might arise from equipment malfunctions or human interference. Even in the presence of multiple seismic events during the GPS observation phase, the proposed method can automatically detect these individual coseismic offset signals. For the case that there are multiple coseismic offsets in the observation data, it is assumed that the changes in first-order differences are caused by seismic events. Parameters are then introduced to model the original GPS

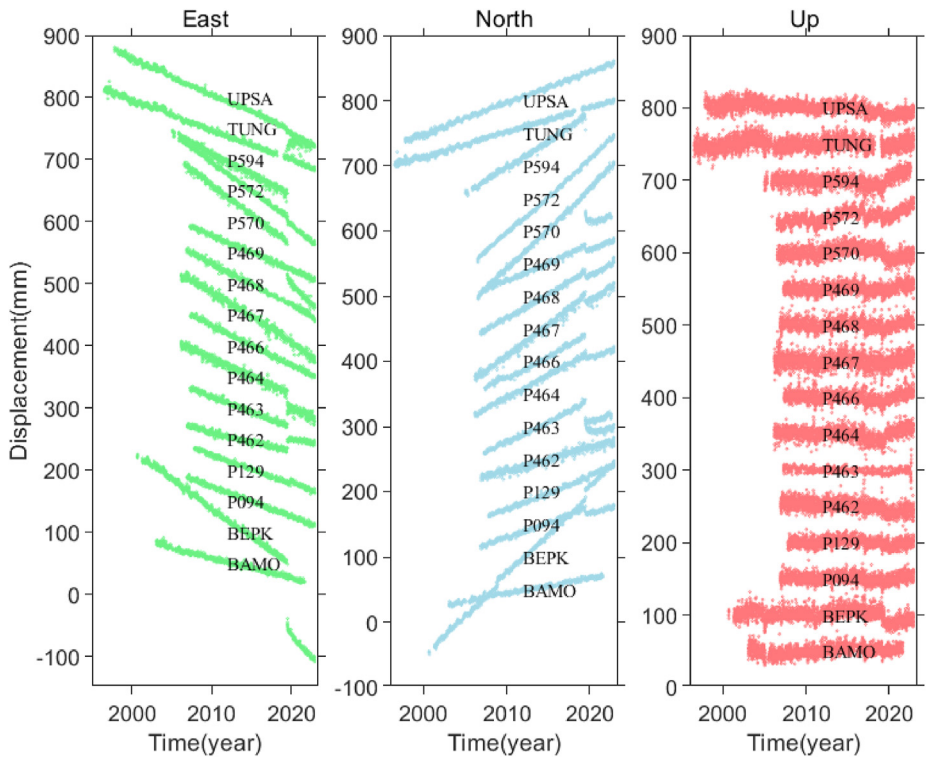

Fig. 5. Complete GPS displacement time series of 16 sites out of all 164 sites. The green dots represent the original time series in the East direction, the light blue dots represent the original time series in the North direction, and the red dots represent the original time series in the Up direction.

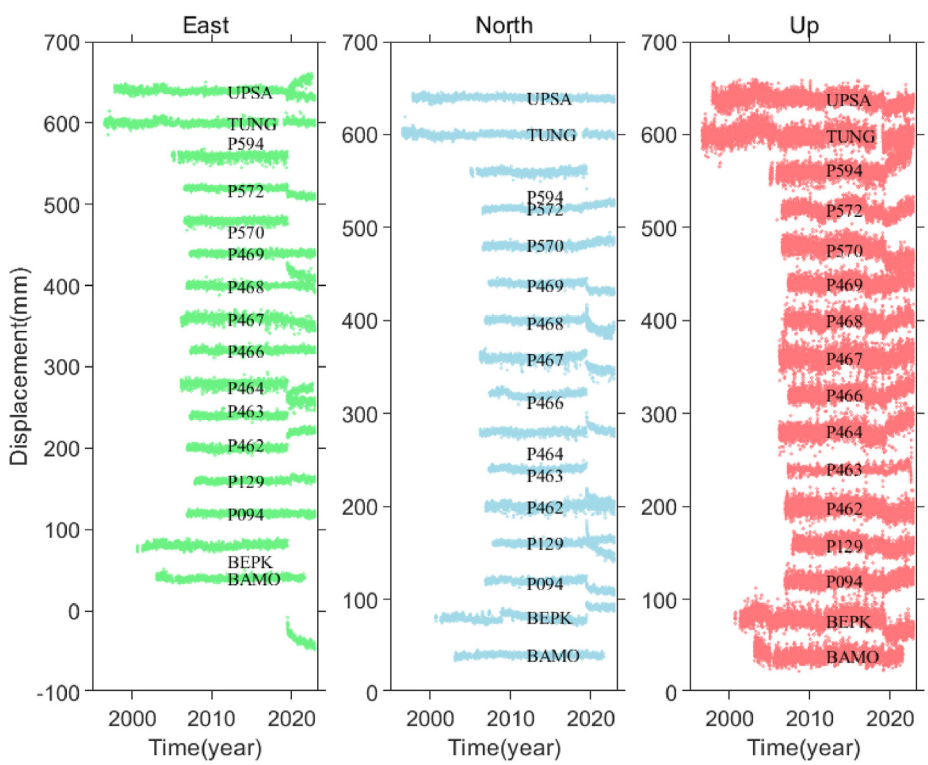

Fig. 6. The detrended GPS displacement time series of 16 sites out of all 164 sites. The green dots represent the detrended time series in the East direction, the light blue dots represent the detrended time series in the North direction, and the red dots represent the detrended time series in the Up direction.

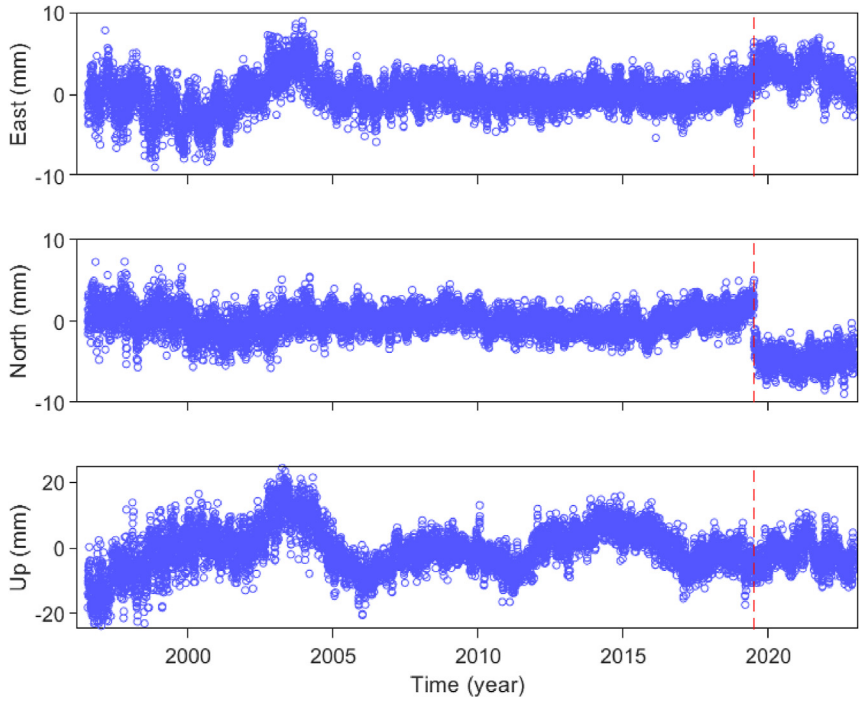

Fig. 7. Stacking of all the detrended GPS coordinate time series of 164 sites. The purple dots represent the observed values, and the red dashed line represents the detected occurrence of the coseismic offset signal.

displacement time series, following by comparative analysis to determine whether the coseismic offsets are indeed earthquake-induced. If they are, the aforementioned method is applied for further detection. Firstly, the parameter τ is determined by one-dimensional grid search, and then the other parameters of the original GPS displacement time series model are re-estimated. Based on the analysis of experimental results, the removal of trend and seasonal terms aims to eliminate effect caused by interseismic velocity, annual and semi-annual periodic oscillation. It is important to consider the variations in observation epochs and data quantities for each site to ensure a consistent and accurate analysis. Considering that the vertical component of GPS observations has higher errors and a lower signal-to-noise ratio compared with the horizontal components, this study primarily focuses on detecting deformation in the E and N directions. Once coseismic offset is detected in any of these directions, the parameters should be simultaneously adjusted for the E, N, and U components, with the U component serving as an auxiliary reference.

The effectiveness of the comprehensive detection method for coseismic offset signals is directly related to the magnitude of the ground displacement recorded by the GPS sites as a result of the earthquake. The observed coseismic offset signal at GPS sites are influenced by various factors, including earthquake magnitude, source depth, epicentral distance, soil geology conditions, source mechanism, topography, and geomorphology. If an earthquake occurs at a shallow depth, the larger magnitude of the earthquake will result in more prominent coseismic offset signals that can be recorded by GPS sites. Furthermore, the parameter determination process is conducted independently for each site, disregarding the spatial relationships among the sites. In fact, there is a spatial relationship between the geometric structure of earthquake surface rupture and the distribution of coseismic displacement. If this spatial correlation can be fully considered, it may be possible to mitigate the interference from other external factors and achieve more reasonable detection results for coseismic offset signals.

7. Conclusions

In this study, we present a new method for automatically detecting coseismic offset signals in GPS coordinate time series. The method effectively detects coseismic offset signals without prior knowledge of the earthquake occurrence time, which can reduce the reliance on earthquake catalog information and the tedious work of GPS coordinate time series analysis to a certain extent. Application to the simulation data and displacement time series of actual GPS sites show the feasible and efficiency of the automatic detection method for coseismic offset signals without additional information.

Conflicts of interest

The authors declare that there is no conflicts of interest.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (No. 42104008, 42204006, 41904031), the Jiangxi Provincial Natural Science Foundation (20232BAB213075), the Key Laboratory for Digital Land and Resources of Jiangxi Province, East China University of Technology (DLLJ202016), and Open Fund of Hubei Luojia Laboratory (No. 230100020, 230100019).

Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.geog.2023.09.001.

References

[1] Y. Bock, D. Agnew, P. Fang, et al., Detection of crustal deformation from the Landers earthquake sequence using continuous geodetic measurements, Nature 361 (6410) (1993) 337–340.

[2] K.M. Larson, J.T. Freymueller, S. Philipsen, Global plate velocities from the global positioning system, J. Geophys. Res. Solid Earth 102 (B5) (1997) 9961–9981.
[3] Y. Bock, D. Melgar, Physical applications of GPS geodesy: a review, Rep. Prog. Phys. 79 (10) (2016) 106801.
[4] L. Métivier, X. Collilieux, D. Lercier, et al., Global coseismic deformations, GNSS time series analysis, and earthquake scaling laws, J. Geophys. Res. Solid Earth 119 (12) (2014) 9095–9109.
[5] F.K. Wyatt, Measurements of coseismic deformation in southern California: 1972–1982, J. Geophys. Res. Solid Earth 93 (B7) (1988) 7923–7942.
[6] L.N. Su, Y. Zhang, Automatic detection and estimation of coseismic and postseismic deformation in GPS time series, Geomatics Inf. Sci. Wuhan Univ. 43 (10) (2018) 1504–1510 [In Chinese with English abstract].
[7] A.R. Amiri-Simkooei, M. Hosseini-Asl, J. Asgari, et al., Offset detection in GPS position time series using multivariate analysis, GPS Solut. 23 (2019) 1–12.
[8] S.D. Williams, Offsets in global positioning system time series, J. Geophys. Res. Solid Earth 108 (B6) (2003).
[9] J. Langbein, Noise in GPS displacement measurements from southern California and southern Nevada, J. Geophys. Res. Solid Earth 113 (B5) (2008).
[10] Y. Fu, J.T. Freymueller, Seasonal and long-term vertical deformation in the Nepal Himalaya constrained by GPS and GRACE measurements, J. Geophys. Res. Solid Earth 117 (B3) (2012).
[11] Y.F. Tian, Z.K. Shen, Extracting the regional common-mode component of GPS station position time series from dense continuous network, J. Geophys. Res. Solid Earth 121 (2) (2016) 1080–1096.
[12] J. Zhang, Continuous GPS measurements of crustal deformation in southern California, University of California, San Diego, 1996.
[13] R. Nikolaidis, Observation of geodetic and seismic deformation with the global positioning system, University of California, San Diego, 2002.
[14] Z.K. Shen, D.D. Jackson, Y. Feng, et al., Postseismic deformation following the Landers earthquake, California, 28 June 1992, Bull. Seismol. Soc. Am. 84 (3) (1994) 780–791.
[15] F. Ming, Research on the GPS coordinate time series analysis, Acta Geod. Cartogr. Sinica 48 (10) (2019) 1340 [In Chinese with English abstract].
[16] T. Van Dam, J. Wahr, P.C.D. Milly, et al., Crustal displacements due to continental water loading, Geophys. Res. Lett. 28 (4) (2001) 651–654.
[17] J.C. Savage, J.L. Svarc, Postseismic deformation associated with the 1992 M ω= 7.3 Landers earthquake, southern California, J. Geophys. Res. Solid Earth 102 (B4) (1997) 7565–7577.
[18] D. Dong, P. Fang, Y. Bock, et al., Spatiotemporal filtering using principal component analysis and Karhunen-Loeve expansion approaches for regional GPS network analysis, J. Geophys. Res. Solid Earth 111 (B3) (2006).
[19] Y. Okada, Surface deformation due to shear and tensile faults in a half-space, Bull. Seismol. Soc. Am. 75 (1985) 1135–1154.

Zhiwei Yang, is a graduate student at East China University of Technology. He mainly engages in GPS coordinate time series analysis and transient deformation detection. E-mail: wy2168429792@163.com